

SYDNEY TECHNICAL HIGH SCHOOL MATHEMATICS DEPARTMENT

Year 12 2 Unit HSC Task No 2 March 2002

Name: _____

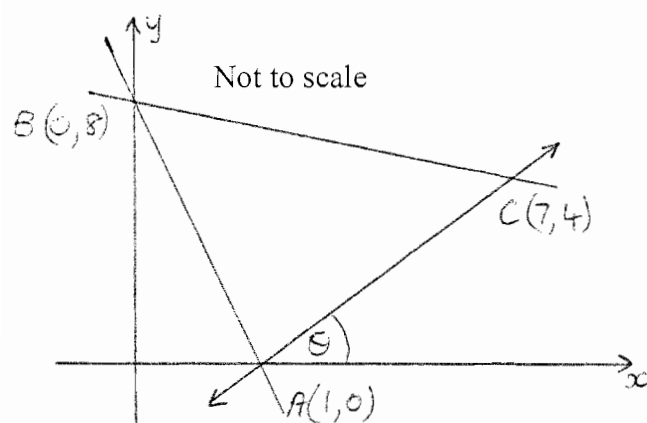
Teacher: _____

Instructions:

- Show all necessary working
- Full marks may not be awarded for incomplete working or poor setting out.
- Approximate marks are indicated
- Hand in this question paper on top of your answer pages.
- TIME ALLOWED : 70 MINUTES

Question 1	Question 2	Question 3	Question 4	Question 5	Total
/10	/10	/10	/10	/10	/50

Question 1



The points A,B,C have coordinates (1,0), (0,8) and (7,4) as shown. The angle between line AC and the x -axis is θ .

Copy this diagram onto your Answer page.

- | | | |
|----|---|---|
| a) | Find the gradient of the line AC | 1 |
| b) | Calculate the size of θ to the nearest degree. | 1 |
| c) | Find the equation of the line AC | 2 |
| d) | Show that $\triangle ABC$ is isosceles | 2 |
| e) | Find the perpendicular distance from B to AC | 2 |
| f) | Find the area of $\triangle ABC$ | 1 |
| g) | Write down the coordinates of a point E such that ABCE is a rhombus | 1 |

Question 2

Give the curve $y = x^3 - 6x^2 + 9x - 5$

- a) Find the stationary points and determine their nature. 4
- b) Sketch the curve for $x \geq -1$ and find the absolute minimum value. 2
- c) For what values of x is: 2

(i) $\frac{dy}{dx} < 0$ (ii) the curve concave up?

- d) Find the equation of the tangent to the curve when $x = 0$ 2

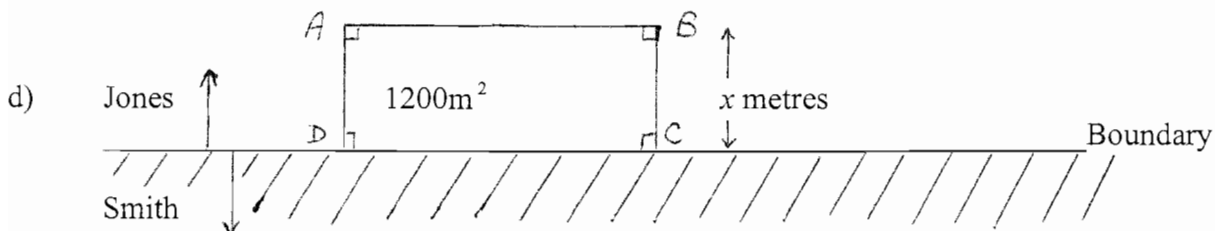
Question 3

- a) The government announces that “ **while unemployment is currently rising its increase will slow.**” 2

Given a mathematical model $y = f(x)$ for unemployment, what does this statement imply for

$\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$?

- b) If $\frac{d^2y}{dx^2} = 6x - 4$ and that $\frac{dy}{dx} = 7$ at $(1, 12)$, find y as a function of x . 2
- c) Find $\int (4x - 6)^9 dx$. 1



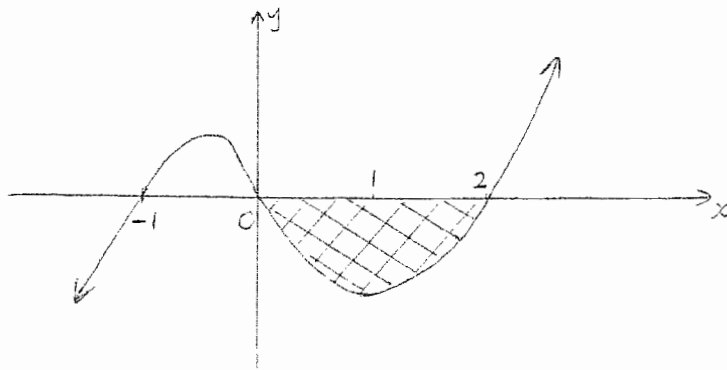
Farmer Jones wishes to fence off a rectangular yard ABCD of area 1200m^2 , as in the figure above, with the side CD against the property of Farmer Smith. Fencing costs \$3 per metre and Smith has agreed to pay for half the cost of side CD. Let \$C be the cost to Jones of fencing the yard and x metres be the length of BC.

- (i) Show that $C = 6x + \frac{5400}{x}$ 1
- (ii) Prove that the minimum cost to Jones for fencing the yard is \$360. 4

Question 4

a) Evaluate $\int_0^3 \sqrt{x+1} \, dx$ 2

- b) The graph of $y = x(x+1)(x-2)$ is shown. 3
Find the shaded area



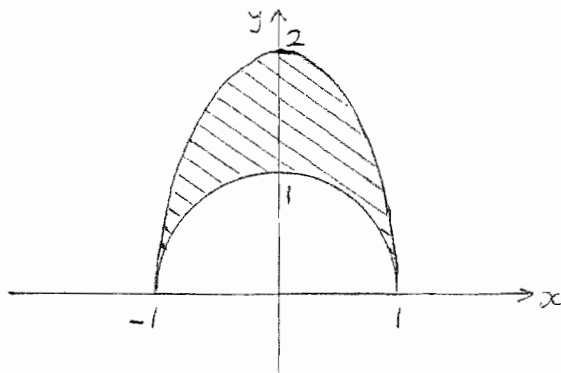
- c) Find the area enclosed between the graphs of $y = x^3$ and $y = 4x$ 5

Question 5

a) Find $\int \frac{3x^3 + 2}{x^2} \, dx$ 2

- b) A cone is generated by rotating about the y -axis the area bounded by the line $y = 2x$, the y -axis and the line $y = 2$. Use integral calculus to find the volume of this cone. 3

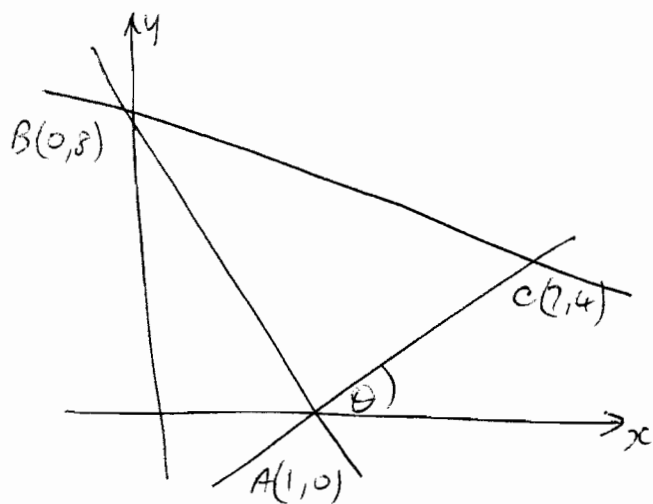
- c) The graphs of $y = \sqrt{1-x^2}$ and $y = 2\sqrt{1-x^2}$ are shown below. 5



Find the volume of the resultant solid when the enclosed region shown above is rotated about the x -axis.

Solutions - 2014 HSC ASS #2 2002

Question 1



$$a) m_{AC} = \frac{4-0}{2-1} = \frac{4}{1} = 4 \quad \text{①}$$

$$b) \tan \theta = \frac{4}{3}$$

$$\therefore \theta \approx 53^\circ \quad \text{①}$$

$$c) y - 0 = \frac{4}{3}(x - 1) \quad \text{①}$$

$$\therefore y = \frac{4}{3}x - \frac{4}{3}$$

$$\text{or } 2x - 3y - 2 = 0 \quad \text{①}$$

$$d) AB = \sqrt{8^2 + 1^2} = \sqrt{65} \quad \text{①}$$

$$BC = \sqrt{7^2 + 4^2} = \sqrt{65} \quad \text{①}$$

$\therefore \triangle ABC$ is isosceles
(2 equal sides)

$$e) 2x - 3y - 2 = 0 \text{ and } (0, 8)$$

$$\therefore p.d. = \frac{|2 \times 0 - 3 \times 8 - 2|}{\sqrt{2^2 + 3^2}} \quad \text{①}$$

$$f) \text{ area} = \frac{1}{2} \times \sqrt{6^2 + 4^2} \times \frac{26}{\sqrt{13}} \\ = \frac{1}{2} \times \frac{\sqrt{4}}{\sqrt{13}} \times \frac{26}{\sqrt{13}} \\ = 26 \text{ u}^2 \quad \text{①}$$

$$g) E(8, -4) \quad \text{①}$$

Question 2

$$a) \text{ S.P.'s when } \frac{dy}{dx} = 0$$

$$\therefore 3x^2 - 12x + 9 = 0 \quad \text{①}$$

$$3(x^2 - 4x + 3) = 0$$

$$3(x-3)(x-1) = 0$$

$$\therefore x = 3 \text{ or } 1 \quad \text{①}$$

x	2.9	3	3.1
$\frac{dy}{dx}$	-	0	+

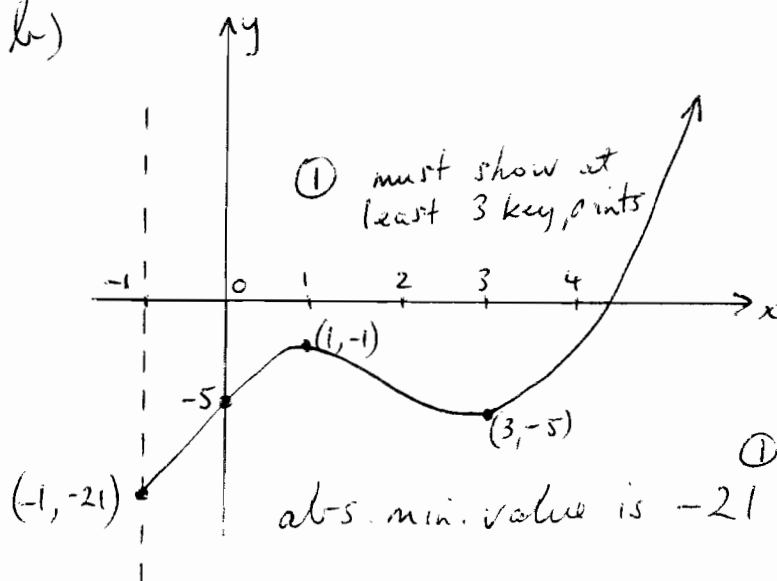
$\searrow$ ①

x	0.9	1	1.1
$\frac{dy}{dx}$	+	0	-

$\nearrow$ ①

$\therefore$ min. T.P. at (3, -5) max T.P. at (1, -1)

b)



$$c) (i) 1 < x < 3 \quad \text{①}$$

$$(ii) \frac{d^2y}{dx^2} = 0, 6x - 12 = 0$$

$$d) \frac{dy}{dx} = 3x^2 - 12x + 9$$

$$\text{When } x=0, \frac{dy}{dx} = m_T = 9 \quad \textcircled{1}$$

$\therefore$ eqn of tangent at $(0, -5)$

$$\text{is } y+5 = 9(x-0)$$

$$\therefore y = 9x - 5 \quad \textcircled{1}$$

Question 3

$$a) \frac{dy}{dx} > 0 \quad \textcircled{1} \text{ and } \frac{d^2y}{dx^2} < 0 \quad \textcircled{1}$$

$$b) \frac{dy}{dx} = 3x^2 - 4x + c$$

$$\text{and } 7 = 3 - 4 + c (\therefore c = 8) \quad \textcircled{1}$$

$$\therefore \frac{dy}{dx} = 3x^2 - 4x + 8$$

$$\therefore y = x^3 - 2x^2 + 8x + k$$

$$\text{and } 12 = 1 - 2 + 8 + k (\therefore k = 5)$$

$$\therefore y = x^3 - 2x^2 + 8x + 5 \quad \textcircled{1}$$

$$c) \frac{(4x-6)^{10}}{10 \times 4} + c = \frac{(4x-6)^{10}}{40} + c \quad \textcircled{1}$$

d) (i) metres of fencing

$$= x + x + \frac{1200}{x} + \frac{600}{x}$$

$$= 2x + \frac{1800}{x} \quad \boxed{1200} \quad x \quad \frac{1200}{x}$$

$\therefore$ cost to Jones

$$= (2x + \frac{1800}{x}) \times \$3/m$$

$$(ii) \text{ min. cost when } \frac{dC}{dx} = 0$$

$$\frac{dC}{dx} = 6 - 5400x^{-2} = 0 \quad \textcircled{1}$$

$$\therefore 6 - \frac{5400}{x^2} = 0$$

$$\therefore 6 = \frac{5400}{x^2}$$

$$\therefore x^2 = \frac{5400}{6} = 900$$

$$\therefore x = 30 \quad \textcircled{1} \text{ (} +ve \text{ only)}$$

x	29	30	31
$\frac{dC}{dx}$	-	0	+

$\therefore$ minimum

$\textcircled{1}$ cost is proved by $x = 30m$,

and cost to Jones is

$$6 \times 30 - \frac{5400}{30} = 180 + 180 \quad \textcircled{1}$$

$$= \$360 \text{ as reqd.}$$

Question 4

$$a) \int_0^3 (x+1)^{1/2} dx = \left[\frac{2}{3} (x+1)^{3/2} \right]_0^3 \quad \textcircled{1}$$

$$= \frac{2}{3} \cdot 4^{3/2} - \frac{2}{3} \cdot 1^{3/2}$$

$$= \frac{2}{3} \times 8 - \frac{2}{3}$$

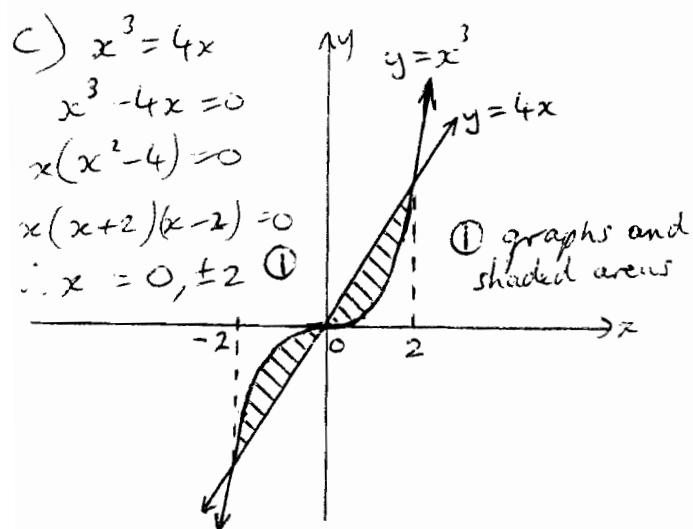
$$= 4 \frac{2}{3} \quad \textcircled{1}$$

$$b) y = x(x^2 - x - 2) = x^3 - x^2 - 2x$$

$$\therefore S.A. = \left| \int_0^2 (x^3 - x^2 - 2x) dx \right| \quad \textcircled{1}$$

$$= \left| \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_0^2 \right| \quad \textcircled{1}$$

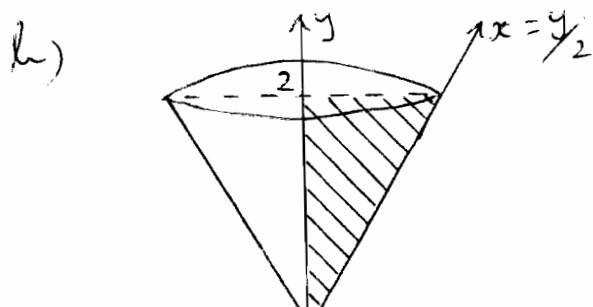
$$= \left| (4 - 2 \frac{2}{3} - 4) - (0 - 0 - 0) \right|$$



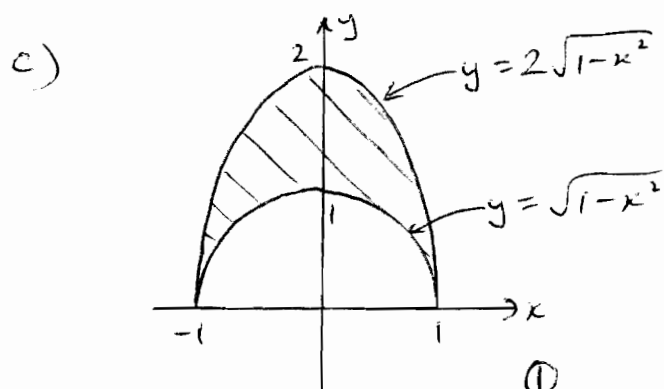
$$\begin{aligned} \text{Area} &= 2 \times \int_0^2 (4x - x^3) dx \quad ① \\ &= 2 \times \left[2x^2 - \frac{x^4}{4} \right]_0^2 \quad ① \\ &= 2 \times \left[(8 - 4) - (0 - 0) \right] \\ &= 8 \text{ u}^2 \quad ① \end{aligned}$$

Question 5

a) $\int \left(\frac{3x^3}{x^2} + \frac{2}{x^2} \right) dx$
 $= \int (3x + 2x^{-2}) dx \quad ①$
 $= \frac{3x^2}{2} + \frac{2x^{-1}}{-1} + C$
 $= \frac{3x^2}{2} - \frac{2}{x} + C \quad ①$



$$\begin{aligned} \text{Vol} &= \pi \int_0^2 \left(\frac{y}{2} \right)^2 dy \quad ① \\ &= \pi \int_0^2 \frac{y^2}{4} dy \\ &= \pi \times \left[\frac{y^3}{12} \right]_0^2 \quad ① \\ &= \pi \times \left(\frac{8}{12} - \frac{0}{12} \right) \\ &= \frac{2\pi}{3} \text{ u}^3 \quad ① \end{aligned}$$



$$\begin{aligned} \text{Vol} &= 2 \times \pi \times \left[\int_0^1 (2\sqrt{1-x^2})^2 dx - \int_0^1 (\sqrt{1-x^2})^2 dx \right] \quad ① \\ &= 2\pi \times \left[\int_0^1 4(1-x^2) dx - \int_0^1 (1-x^2) dx \right] \\ &= 2\pi \times \int_0^1 (4 - 4x^2 - 1 + x^2) dx \\ &= 2\pi \times \int_0^1 (3 - 3x^2) dx \quad ① \\ &= 2\pi \times \left[3x - x^3 \right]_0^1 \quad ① \\ &= 2\pi \times [(3 - 1) - (0 - 0)] \\ &= 2\pi \times 2 \\ &= 4\pi \text{ u}^3 \quad ① \end{aligned}$$